

Self-avoiding walks in Anderson localisation

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Geboren am 21. November 2004 in Bologna

20th September 2026

Bachelor's Thesis Mathematics

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Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 [And58] to explain the quantum mechanical effects of disorder in materials. The model considers a d -dimensional grid, in which each site represents an atom. Anderson discusses the behaviour of electrons in dirty crystals, which is the quantum mechanical analogue of a random walk in a random environment. From a heuristical point of view, we consider electrons that jump from atom to atom, subject to an external random potential that influences their motion. When the disorder is strong enough, these particles become trapped in a specific region and can no longer spread to generate electric conduction. This complete absence of quantum transport is called Anderson localisation, and it essentially turns the material into an insulator. This is in sharp contrast with the behaviour of electrons in ideal, perfectly ordered crystals, which always act as conductors.

In the last decades, the study of random operators has become an important field of mathematical physics, leading to a tremendous amount of research activity and many rigorous mathematical results.

The two important methods to prove Anderson localization are the *multiscale analysis* (MSA), developed by Fröhlich and Spencer [FS83], and the *fractional moment method* (FMM) by Aizenman and Molchanov [AM93]. Even though the multiscale analysis method has lead to a multitude of results on spectral and dynamical localization for a wide range of models, in this thesis we will focus on the fractional moments method, which has provided a simple perspective on localization and has enabled exponentially decaying bounds on expectation values of various propagation kernels.

Structure of the thesis In Chapter 1, we introduce the Anderson model, focusing on the properties of its components, its spectrum, and the main results known to date.

In Chapter 2, we define spectral and dynamical localisation. We also

define exponential dynamical localisation and show how it implies that all moments of the position operator remain bounded in time.

In Chapter 3, we introduce the Green function as the resolvent of the model and explore how the exponential decay of its fractional power leads to localisation. Specifically, we rely on simple analytical tools, such as the resolvent identity of linear operators and the resulting a priori bound.

In Chapter 4, we provide the preliminary definitions for self-avoiding walk (SAW) methods and obtain the main results of this thesis using a new representation for the Green's function.

The final chapter offers a comparison between the standard model with nearest-neighbor jumps and a model with long-range jumps. We define the fractional discrete Laplacian and basically follow the proof structure of the previous Chapters in order to obtain similar results but with a polynomial decay.

Prerequisites It is useful to have some familiarity with measure theory, basic probabilistic concepts such as independence and continuous distributions, foundations of the theory of linear operators in Hilbert spaces, up to a general idea of the consequences of the spectral theorem for self adjoint operators.

Chapter 1

The Anderson Model

Transport phenomena in a disordered material are often described via discrete random Schrödinger operator. In quantum mechanics, it corresponds to the classical total energy decomposing into kinetic and potential energy. On the lattice $\mathbb{Z}^d$, it takes the form of an infinite random matrix called the *Hamiltonian*:

$$H_\omega = -\Delta + \lambda V_\omega. \quad (1.0.1)$$

where $-\Delta$ is a deterministic matrix, V_ω is a diagonal matrix with random entries and λ is the disorder parameter.

This operator acts on the Hilbert space

$$\ell^2(\mathbb{Z}^d) = \{u : \mathbb{Z}^d \rightarrow \mathbb{C} : \sum_{x \in \mathbb{Z}^d} |u(x)|^2 < \infty\}$$

with inner product $\langle u, v \rangle = \sum_n \overline{u(x)}v(x)$.

In quantum mechanics, a physical state describing a single particle (such as an electron) is represented by a normalized wave function $\psi \in \ell^2(\mathbb{Z}^d)$:

$$\|\psi\|_{\ell^2}^2 = \sum_{x \in \mathbb{Z}^d} |\psi(x)|^2 = 1.$$

The quantity $P(x) = |\psi(x)|^2$ defines a probability mass function on the lattice $\mathbb{Z}^d$: it represents the probability of finding the particle at site $x \in \mathbb{Z}^d$. This functions are the solutions to the time-evolving Schrödinger operator $H_\omega \phi(t) = i\partial_t \phi(t)$. Because the time evolution operator e^{-itH_ω} is unitary on

$\ell^2(\mathbb{Z}^d)$, the total probability is strictly conserved at all times $t \in \mathbb{R}$:

$$\|\psi(t)\|_{\ell^2}^2 = \|e^{-itH_\omega}\psi_0\|_{\ell^2}^2 = \|\psi_0\|_{\ell^2}^2 = 1. \quad (1.0.2)$$

In particular, if the particle is initially placed at a specific site $x \in \mathbb{Z}^d$ with $\psi(0) = \delta_x$, the transition probability to find the particle at site $y \in \mathbb{Z}^d$ at time t is given by the squared transition amplitude:

$$P(x \rightarrow y; t) = |e^{-itH_\omega}(x, y)|^2. \quad (1.0.3)$$

As we'll see later on, under certain conditions on the disorder parameter, P tends to 0 faster than any polynomial at all t for $|x - y| \rightarrow \infty$.

1.1 The Hamiltonian

The Discrete Laplacian is the discrete analogue of the usual Laplacian $(\sum_j \partial^2 / \partial x_j^2)$. For the functions $u \in \ell^2(\mathbb{Z}^d)$ it is defined as:

$$-\Delta u(x) := \sum_{y \in \mathbb{Z}^d, |x-y|=1} u(x) - u(y) \quad \text{for } x \in \mathbb{Z}^d. \quad (1.1.1)$$

where $|\cdot|$ denotes the ℓ^2 norm. For each site x , there are $2d$ possible y such that $|x - y| = 1$. Thus, we can define an infinite matrix such that:

$$(-\Delta)(x, y) = \begin{cases} 2d & \text{if } x = y \\ -1 & \text{if } |x - y| = 1 \\ 0 & \text{otherwise} \end{cases}$$

The discrete Laplacian represents the kinetic term and connects neighbour sites on the lattice, such that the electron can transition from one to another.

For $u \in \ell^2(\mathbb{Z}^d)$, $-\Delta u$ is summable in ℓ^2 , bounded, and self-adjoint. In fact we can show:

$$\begin{aligned}
\sum_{x \in \mathbb{Z}^d} |-\Delta u(x)|^2 &= \sum_{x \in \mathbb{Z}^d} \left| \sum_{y \in \mathbb{Z}^d, |x-y|=1} (u(x) - u(y)) \right|^2 \\
&\leq \sum_{x \in \mathbb{Z}^d} \left(2d \sum_{|x-y|=1} |u(x) - u(y)|^2 \right) \\
&\leq 2d \sum_{x \in \mathbb{Z}^d} \left(\sum_{|x-y|=1} (|u(x)|^2 + |u(y)|^2 + 2|u(x)u(y)|) \right) \\
&\leq 2d \sum_{x \in \mathbb{Z}^d} \left(2 \sum_{|x-y|=1} (|u(x)|^2 + |u(y)|^2) \right) \\
&\leq 2d \cdot 2d \cdot 4 \|u\|_2^2 < \infty
\end{aligned}$$

where we used the bound over the number of neighbors and Cauchy Schwarz, then the Young's inequality and in the last step the invariance under translation of the ℓ^2 norm.

The boundedness comes from:

$$\frac{\|\Delta u\|_{\ell^2}^2}{\|u\|_{\ell^2}^2} \leq 16d^2 \implies \|\Delta\| = \sup_{u \neq 0} \frac{\|\Delta u\|_{\ell^2}}{\|u\|_{\ell^2}} \leq 4d.$$

To prove self-adjointness, we only need to show symmetry (since $-\Delta$ is bounded). For $u, v \in \ell^2(\mathbb{Z}^d)$ and $y \in \mathbb{Z}^d$:

$$\begin{aligned}
\langle u, -\Delta v \rangle &= \sum_{x \in \mathbb{Z}^d} \overline{u(x)} (-\Delta v(x)) = \sum_{x \in \mathbb{Z}^d} \overline{u(x)} \left(\sum_{|x-y|=1} v(x) - v(y) \right) \\
&= \sum_{x \in \mathbb{Z}^d} \left(\overline{u(x)} 2dv(x) - \sum_{|x-y|=1} \overline{u(x)} v(y) \right) \\
&= \sum_{x \in \mathbb{Z}^d} 2d \overline{u(x)} v(x) - \sum_{x \in \mathbb{Z}^d} \sum_{|x-y|=1} \overline{u(y)} v(x) \\
&= \sum_{x \in \mathbb{Z}^d} v(x) \left(\sum_{|x-y|=1} \overline{u(x)} - \overline{u(y)} \right) \\
&= \sum_{x \in \mathbb{Z}^d} v(x) \overline{(-\Delta u(x))} = \langle -\Delta u, v \rangle.
\end{aligned}$$

The spectrum of a bounded, self-adjoint operator is real-valued, and in this case equal to the closed interval $[0, 4d]$. In this range, we find the levels of kinetic energy that an electron owns. In order to understand this result better, we consider the discrete Fourier transform:

$$\mathcal{F}: \ell^2(\mathbb{Z}^d) \longrightarrow L^2([0, 2\pi)^d)$$

such that:

$$(\mathcal{F}u)(k) := \lim_{N \rightarrow \infty} (2\pi)^{-\frac{d}{2}} \sum_{|x| \leq N} u(x) e^{-ik \cdot x}, \quad k \in [0, 2\pi)^d.$$

We consider now calculations for $u \in \ell^1(\mathbb{Z}^d)$. For ℓ^1 is a dense subset of ℓ^2 and $\mathcal{F}$ unitary, we can extend the results by continuity for all functions $\ell^2(\mathbb{Z}^d)$.

$$\begin{aligned} (\mathcal{F}(-\Delta u))(k) &= (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} (-\Delta u)(x) e^{-ik \cdot x} \\ &= (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} \left(\sum_{|y-x|=1} (u(x) - u(y)) \right) e^{-ik \cdot x} \\ &= (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} \left(2d u(x) - \sum_{|y-x|=1} u(y) \right) e^{-ik \cdot x} \\ &= 2d (\mathcal{F}u)(k) - (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} \sum_{|y-x|=1} u(y) e^{-ik \cdot x} \\ &= 2d (\mathcal{F}u)(k) - (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} u(y) \sum_{|m|=1} e^{-ik \cdot (y+m)} \quad (\text{for } m = x - y) \\ &= 2d (\mathcal{F}u)(k) - \left((2\pi)^{-\frac{d}{2}} \sum_{y \in \mathbb{Z}^d} u(y) e^{-ik \cdot y} \right) \sum_{|m|=1} e^{-ik \cdot m} \\ &= 2d (\mathcal{F}u)(k) - (\mathcal{F}u)(k) \sum_{j=1}^d (e^{-ik_j} + e^{ik_j}) \\ &= \left(2 \sum_{j=1}^d (1 - \cos(k_j)) \right) (\mathcal{F}u)(k). \end{aligned}$$

We obtain the multiplication operator:

$$\mathcal{F}\Delta\mathcal{F}^{-1} = 2 \sum_{j=1}^d (1 - \cos(k_j))$$

on $L^2([0, 2\pi)^d)$ in the variable $k = (k_1, \dots, k_d)$. We notice that the Laplacian doesn't have eigenvalues in $\ell^2(\mathbb{Z}^d)$. Another similarity of Δ with the continuum Laplacian is that it has plane waves as generalized eigenfunctions. To see this, let $k \in [0, 2\pi)^d$ and set

$$\phi_k(x) := e^{-ix \cdot k}$$

While $\phi_k \notin \ell^2(\mathbb{Z}^d)$, Δ acts on it as

$$\begin{aligned} (\Delta\phi_k)(x) &= \sum_{|y|=1} (e^{-in \cdot k} - e^{-i(x+y) \cdot k}) \\ &= 2d\phi_k(x) - \left(\sum_{j=1}^d 2 \cos(k_j) \right) \phi_k(x) = \left(2 \sum_{j=1}^d (1 - \cos(k_j)) \right) \phi_k. \end{aligned}$$

Thus ϕ_k is a bounded generalized normalized eigenfunction of Δ to the spectral value $2 \sum_{j=1}^d (1 - \cos(k_j))$. thus, the spectrum is continuous and bounded in $[0, 2(2d)]$.

The Random Potential is expressed as the multiplication operator $V_\omega(x) = \omega_x$. It is a diagonal matrix, where $\omega = (\omega_x)_{x \in \mathbb{Z}^d}$ is the set of the independent identically distributed real-valued random entries. Physically, V_ω represents the random electric potential created by nuclei at the sites $x \in \mathbb{Z}^d$.

Definition 1.1.1 (i.i.d. random variables). Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a general probability space.

A family of random variables $(\omega_x)_{x \in \mathbb{Z}^d}$ defined on $(\Omega, \mathcal{A}, \mathbb{P})$ is **identically distributed** if there exists a Borel probability measure μ on $\mathbb{R}$ such that $\mathbb{P}(\omega_x \in A) = \mu(A)$ for all $x \in \mathbb{Z}^d$ and all Borel sets $A \subset \mathbb{R}$.

The family $(\omega_x)_{x \in \mathbb{Z}^d}$ is **independent** if

$$\mathbb{P}(\omega_{x_1} \in A_1, \dots, \omega_{x_l} \in A_l) = \prod_{j=1}^l \mathbb{P}(\omega_{x_j} \in A_j) = \prod_{j=1}^l \mu(A_j)$$

for every finite subset of distinct indices $\{x_1, \dots, x_l\} \subset \mathbb{Z}^d$ and arbitrary Borel sets $A_1, \dots, A_l \subset \mathbb{R}$.

In this thesis we assume that ω_x are uniformly distributed in the real closed interval $[-1, 1]$.

We consider then the probability space as the following infinite product space:

$$(\Omega, \mathcal{A}, \mathbb{P}) = \bigotimes_{x \in \mathbb{Z}^d} ([-1, 1], \mathcal{B}([-1, 1]), \mu),$$

where $\mathcal{B}([-1, 1])$ is the Borel σ -algebra and $d\mu(\omega_x) = \rho(\omega_x)d\omega_x$ where the continuous density is given by $\rho = \frac{1}{2}\chi_{[-1, 1]}$.

It is possible to generalise the results to all absolutely continuous distributions where ρ has compact bounded support.

As a diagonal operator, the spectrum of V_ω consists of the random entries ω_x , each corresponding to an eigenvector given by the Dirac delta $\delta_x \in \ell^2(\mathbb{Z}^d)$.

As for the discrete laplacian, the random term is summable in $\ell^2(\mathbb{Z}^d)$, bounded and self-adjoint. Self-adjointness of the maximal multiplication operator by a real-valued function is a fact. For definition V_ω is bounded in the closed interval $[-1, 1] \subset \mathbb{R}$, and since $\|V_\omega u\|_{\ell^2}^2 \leq \|u\|_{\ell^2}^2$ it is also summable in $\ell^2(\mathbb{Z}^d)$.

The Disorder Parameter is represented by $\lambda > 0$. The value of λ is a great indicator for localization, which makes it one of the central objects of interest of this thesis. It allows us to adjust the strength of the random disorder with respect to the deterministic part.

Let's consider the two extreme cases:

- For $\lambda = 0$ the Hamiltonian loses the random term and only considers the Laplacian. Without disorder, the material behaves like a perfect crystal without localization. The state functions are continuous plane waves and the spectrum is also continuous between $[0, 4d]$.

- For $\lambda \gg 1$ the kinetic hopping term becomes negligible compared to the random potential, so that $H_\omega \approx \lambda V_\omega$. In this scenario electrons are unable to hop between adjacent sites, and the material behaves as an insulator. In this limit, the system exhibits trivial localization: the eigenfunctions are localized Dirac deltas δ_x , and the spectrum is confined to the compact interval $[-\lambda, \lambda]$

1.2 The spectrum

The next goal is to determine the spectrum of H_ω . Because the random potential entries $(\omega_x)_{x \in \mathbb{Z}^d}$ are independent and identically distributed, the random Hamiltonian $H_\omega = -\Delta + \lambda V_\omega$ is an ergodic family of bounded self-adjoint operators. A standard consequence of ergodic operator theory is that the spectrum of H_ω is almost surely deterministic: there exists a closed, non-random set $\Sigma \subset \mathbb{R}$ such that

$$\sigma(H_\omega) = \Sigma \quad \text{for } \mathbb{P}\text{-almost all } \omega \in \Omega. \quad (1.2.1)$$

More specifically, as proven in [Sto11, Theorem 2], the almost sure spectrum can be determined explicitly as the Minkowski sum of the spectrum of the kinetic energy operator and the support of the random potential distribution:

$$\Sigma = \sigma(-\Delta) + \lambda \operatorname{supp}(\mu). \quad (1.2.2)$$

Recall from Section 1.1 that under our convention for the discrete Laplacian, its spectrum is the band $\sigma(-\Delta) = [0, 4d]$. The random potential V_ω has i.i.d. entries distributed uniformly on $[-1, 1]$, making the almost sure spectrum of λV_ω the interval $[-\lambda, \lambda]$. Consequently, the almost sure spectrum of H_ω is given by the compact interval

$$\sigma(H_\omega) = [0, 4d] + [-\lambda, \lambda] = [-\lambda, 4d + \lambda]. \quad (1.2.3)$$

This confirms that the spectrum is a bounded subset of $\mathbb{R}$.

Before characterising the physical meaning of this spectrum, we need some spectral theory preliminaries.

Let μ be a Borel measure on $\mathbb{R}$. Consider now $P = \{x \in \mathbb{R} \mid \mu(\{x\}) \neq 0\}$ the

set of **pure points** of μ . We define the pure point measure as

$$\mu_{pp}(X) = \mu(P \cap X)$$

for $X \subset \mathbb{R}$. In particular, there exists a continuous measure μ_c such that $\mu_c(\{x\}) = 0$ for all $x \in \mathbb{R}$ and $\mu_c = \mu - \mu_{pp}$. Since $\mu_{pp}(P^c) = 0$ and $\mu_c(P) = 0$, the two measures are concentrated on disjoint sets, and thus mutually singular.

One can apply this same division also for spectral measures:

Definition 1.2.1. Let $A : \mathcal{H} \rightarrow \mathcal{H}$ be a bounded, self-adjoint operator. Let $\phi \in \mathcal{H}$ and $f \in C(\sigma(A))$. Then there is a unique measure μ_ϕ on the compact set $\sigma(A)$ with $\langle \phi, f(A)\phi \rangle = \int_{\sigma(A)} f(\xi) d\mu_\phi(\xi)$. The measure μ_ϕ is the *spectral measure associated with the vector ϕ* .

Thus, by the spectral theorem for self-adjoint operators (see [Rud87] and [RS80]), the Hilbert space (for example also $\ell^2(\mathbb{Z}^d)$) decomposes uniquely into an orthogonal direct sum of invariant subspaces:

$$\mathcal{H} = \mathcal{H}_{pp} \oplus \mathcal{H}_c, \quad (1.2.4)$$

where:

- $\mathcal{H}_{pp} = \{\phi \mid \mu_\phi \text{ is pure point}\};$
- $\mathcal{H}_c = \{\phi \mid \mu_\phi \text{ is continuous}\}$

In the case of self-adjoint bounded operators, the restriction to these subspaces defines two, not necessarily disjoint, subsets of the spectrum:

$$\sigma(H_\omega) = \sigma_{pp}(H_\omega) \cup \sigma_c(H_\omega) \quad (1.2.5)$$

where $\sigma_{pp}(H_\omega) = \overline{\{\lambda \in \mathbb{R} \mid \lambda \text{ is an eigenvalue of } H_\omega\}}$.

From a physical perspective, states in $\mathcal{H}_{pp}$ correspond to bound states that remain confined to finite spatial regions for all times, whereas states in the continuous subspace $\mathcal{H}_c$ represent scattering states associated with transport and wave-packet spreading. This heuristic meaning is mathematically expressed by the RAGE theorem (named after Ruelle, Amrein, Georgescu, and Enss):

Theorem 1.2.2 (RAGE Theorem). *Let H be a self-adjoint Schrödinger operator on $\ell^2(\mathbb{Z}^d)$. Then:*

(i) $\phi \in \mathcal{H}_{\text{pp}}$ if and only if

$$\lim_{R \rightarrow \infty} \sup_{t \in \mathbb{R}} \|\chi_{\{|x| > R\}} e^{-itH} \phi\| = 0. \quad (1.2.6)$$

(ii) $\psi \in \mathcal{H}_{\text{c}}$ if and only if for all $R > 0$,

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \|\chi_{\{|x| \leq R\}} e^{-itH} \psi\|^2 dt = 0. \quad (1.2.7)$$

Here, $\chi_{\{|x| \leq R\}}$ and $\chi_{\{|x| > R\}}$ denote the projection operators corresponding to the characteristic functions of the ball $\{x \in \mathbb{Z}^d : |x| \leq R\}$ and its complement, respectively.

In terms of this classification, Anderson localization corresponds to the complete suppression of continuous spectrum, ensuring that $\sigma(H_\omega) = \sigma_{\text{pp}}(H_\omega)$ almost surely.

1.3 Main results and Open Problems

There exist many studies on the Anderson localisation model, led by both physicist and mathematicians. Various discoveries have been made in the past years and yet deep challenges remain.

The main result of this thesis states that if $s \in (0, 1)$ and $\lambda > \lambda_0(s)$

$$\mathbb{E} \left(\left| \langle \delta_x, \frac{1}{(H_\omega - z)} \delta_y \rangle \right|^s \right) \leq C e^{-\mu|x-y|} \quad (1.3.1)$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$, for all $x, y \in \mathbb{Z}^d$ and $C < \infty, \mu > 0$ where $\lambda_0 = \lambda_{\text{And}}$ is a fixed value proposed by P.W. Anderson ([And58]), and rigourously proven later on by J. Schenker ([Sch15]) using self-avoiding walks.

As we can see, the resolvent is well-defined only if z is not an element of $\sigma(H_\omega)$. The reason why the *fractional moments method* was so revolutionary, is that by averaging the fractional power of the resolvent, we avoid this divergences and the bound holds uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$.

Originally, absence of conductivity was proved using the squared-power instead of the fractional, which creates problems at zero and needs complicated analytical work such as the MSA method.

The intuition behind the use of fractional methods, can be trivially understood from this integral, which we'll encounter in the next proofs (for example

see 3.2.3):

$$\int_{\Sigma} \frac{1}{|x|^s} dx = \left[\frac{x^{1-s}}{1-s} \right]_{\Sigma} \begin{cases} < \infty & \text{for } s \in (0, 1) \\ = \infty & \text{for } s \geq 1 \text{ in } x = 0 \end{cases}$$

where $\Sigma \subset \mathbb{R}$ closed.

Very important results concern the value of λ . Based on the dimension of the grid, we know the following.

- $d = 1$.

For the one-dimensional grid, it has been proven, that localisation holds for all $\lambda > 0$ [GMP77].

- $d = 2$.

For $d = 2$ it has been numerically established that localization persists at arbitrarily small disorder.

- $d \geq 3$.

For $d \geq 3$, to find the exact value of λ , until which the material is still a conductor, is an open problem. The best that we can prove is given in this thesis and only states, that localization shows from a specific λ_0 on (see definition in Theorem 3.2.1).

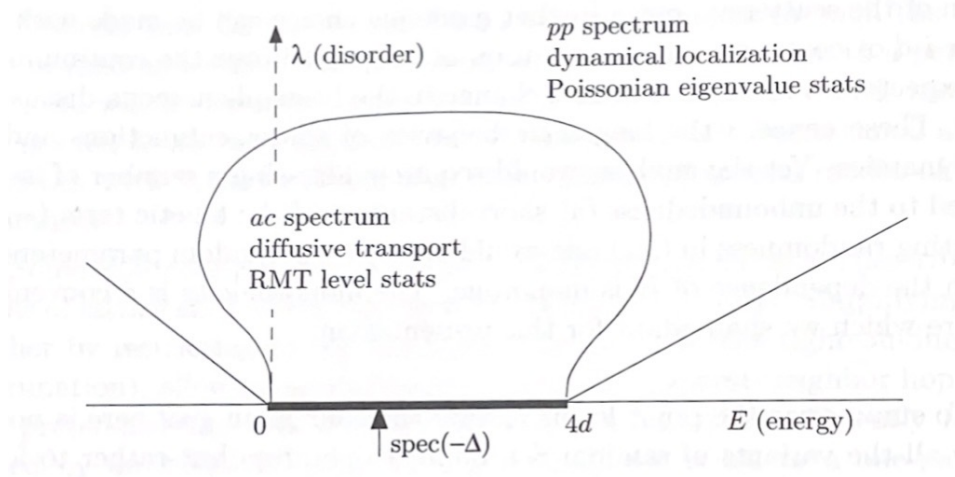


Figure 1.1: For small disorder localization we can still find continuous spectrum

Chapter 2

Localisation properties

We understood what localisation means from an heuristical point of view. In the previous Chapter, it came out as a spectral property of the Hemiltonian. In this section, we also introduce dynamical properties for localization, together with the implications between them.

2.1 Spectral Localisation

Definition 2.1.1 (Spectral Localisation). We say that $H = H_\omega$ exhibits **spectral localisation** if it has pure point spectrum almost surely. That is

$$\sigma(H_\omega) = \sigma_{pp}(H_\omega) \quad \text{and} \quad \sigma_c(H_\omega) = \emptyset \quad (2.1.1)$$

for almost every realization of $\omega \in \Omega$.

2.1.1 Absence of transport

$$\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t) = \| |x| e^{-itH_\omega} P_{H_\omega \in [a,b]} \phi_0 \|^2 \quad (5)$$

If the electrons with energies in $[a, b]$ move ballistically with average velocity v_{av} , then roughly $x(t) \sim v_{av}t$ for large times. Hence $\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)$ will be proportional to t^2 for large t .

On the other hand, if the electrons in the energy interval $[a, b]$ are localized, then it should be natural to expect that $\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)$ is bounded uniformly in t . It is known (Simon, 1990) that pure point spectrum implies absence of ballistic motion,

$$\lim_{t \rightarrow \infty} \frac{\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)}{t^2} = 0$$

for all compactly supported initial conditions ϕ_0 , as soon as the spectrum in $[a, b]$ is pure point. However, del Rio et al. (1996) constructed examples of (non-random) one-dimensional Schrödinger operators with pure point spectrum for all energies and exponentially localized eigenfunctions for which

$$\limsup_{t \rightarrow \pm\infty} \frac{\langle x^2 \rangle_{\phi_0}(t)}{t^{2-\delta}} = \infty \quad \text{for all } \delta > 0, \quad (6)$$

for a large class of compactly supported initial conditions ϕ_0 . *Il fallimento risiede nella libertà delle costanti $C_{\omega,n}$: se $C_{\omega,n}$ cresce incontrollatamente con n , le code delle autofunzioni possono sovrapporsi su scale spaziali arbitrariamente grandi.*

2.2 Dynamical Localisation

Definition 2.2.1 (Dynamical localisation). We say that H_ω shows dynamical localisation if for all $p > 0$,

$$\sup_{t \in \mathbb{R}} \sum_{x \in \mathbb{Z}^d} |x|^p |e^{-itH_\omega}(0, x)|^2 < \infty \quad (2.2.1)$$

for almost every realization of $\omega \in \Omega$.

Definition 2.2.2 (Exponential localisation). We say that H_ω shows exponential localisation if

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |e^{-itH_\omega}(0, x)|^2 \right) \leq C e^{-\mu|x|} \quad (2.2.2)$$

for $x \in \mathbb{Z}^d$

Remark 2.2.3. Dynamical localisation implies that all moments of the distance between the two sites are bounded uniformly in time (recall that we are working in an infinite grid, and with translation invariance). From (2.2.1) we already notice that the probability of transitioning from 0 to x goes to zero faster than any polynomial, but (2.2.2) explicits this directly with an exponential bound. In fact, the two definitions are strongly related and we can prove the following implication.

Proposition 2.2.4. *Exponential localization implies that all moments of the position operator are bounded in time for almost every realisation of $\omega \in \Omega$, i.e., dynamical localisation.*

Proof. To see how (2.2.1) follows from (2.2.2), we argue

$$\begin{aligned} \mathbb{E} \left(\sup_t \sum_x |x|^p |e^{-itH_\omega}(0, x)|^2 \right) &\leq \sum_x |x|^p \mathbb{E} \left(\sup_t |e^{-itH_\omega}(0, x)|^2 \right) \\ &\leq C \sum_x |x|^p e^{-\mu|x|} \\ &< \infty. \end{aligned}$$

□

Remark 2.2.5. The reason why we obtain dynamical localization *almost surely* (rather than for every single realization of the potential) is that our bounds apply to the expected value over the disorder, rather than directly to the norm. An expectation bound alone only tells you about the average behavior (a single realization could still be huge). But if you have expectation bounds on infinitely many random variables that shrink fast enough, then "being large" becomes such a rare event across the whole sequence that (with probability 1) it can only happen finitely often. There are some standard proof using Borel-Cantelli or the Markov inequality.

Lemma 2.2.6. *Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space. Let $h : \mathbb{R} \rightarrow [0, \infty)$ be a monotone, non-negative, growing and measurable function, $A : \Omega \rightarrow \mathbb{R}$ a random variable. Then, if $\mathbb{E}[h(A)] < \infty$, $A < \infty$ $\mathbb{P}$ -almost surely.*

Proof. For every $M > 0$, Markov's inequality gives $\mathbb{P}(A \geq M) \leq \frac{\mathbb{E}[h(A)]}{M}$. Define $\Omega_0 := \{\omega \in \Omega \mid A(\omega) = \infty\}$. Since $\Omega_0 \subseteq \{A \geq M\}$ for every M , we get $\mathbb{P}(\Omega_0) \leq \frac{\mathbb{E}[h(A)]}{M} \rightarrow 0$ as $M \rightarrow \infty$, so $\mathbb{P}(\Omega_0) = 0$. □

2.3 From dynamical to spectral localisation

Proposition 2.3.1. *Suppose that dynamical localization in the form*

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |\langle e_j, e^{-it h_\omega} \chi_I(h_\omega) e_k \rangle| \right) \leq C e^{-\mu|j-k|} \quad (2.3.1)$$

holds. Then h_ω almost surely has pure point spectrum in I .

Proof. For a discrete Schrödinger operator $h = h_0 + V$ in $\ell^2(\mathbb{Z}^d)$ let $P_{\text{cont}}(h)$ be the projection onto its continuous spectral subspace. Then the RAGE-

Theorem says that for every $\psi \in \ell^2(\mathbb{Z}^d)$,

$$\|P_{\text{cont}}(h)\chi_I(h)\psi\|^2 = \lim_{R \rightarrow \infty} \lim_{T \rightarrow \infty} \int_0^T \frac{dt}{T} \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h)\psi\|^2. \quad (2.3.2)$$

If ψ has finite support, say $\text{supp } \psi \subset \{|x| \leq r\}$, then

$$\begin{aligned} \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h)\psi\|^2 &\leq \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \chi_{\{|x| \leq r\}}\|^2 \|\psi\|^2 \\ &\leq \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \chi_{\{|x| \leq r\}}\| \|\psi\|^2 \\ &\leq \sum_{|x| \geq R, |y| \leq r} |\langle e_x, e^{-ith} \chi_I(h) e_y \rangle| \|\psi\|^2, \end{aligned}$$

where dropping a square is allowed as $\|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \chi_{\{|x| \leq r\}}\| \leq 1$.

Taking expectations in (2) implies, after using Fatou and Fubini,

$$\mathbb{E} (\|P_{\text{cont}}(h_\omega) \chi_I(h_\omega) \psi\|^2) \leq \lim_{R \rightarrow \infty, T \rightarrow \infty} \int_0^T \frac{dt}{T} \sum_{|x| \geq R, |y| \leq r} \mathbb{E} \left(|\langle e_x, e^{-ith_\omega} \chi_I(h_\omega) e_y \rangle| \right) \|\psi\|^2. \quad (2.3.3)$$

By (2.4) we have $\mathbb{E} (|\langle e_x, e^{-ith_\omega} \chi_I(h_\omega) e_y \rangle|) \leq C e^{-\mu|x-y|}$ uniformly in t , which bounds the right hand side of (2.6) by

$$\leq \lim_{R \rightarrow \infty} \tilde{C} \sum_{|x| \geq R, |y| \leq r} e^{-\mu|x-y|} = 0.$$

We conclude that $P_{\text{cont}}(h_\omega) \chi_I(h_\omega) \psi = 0$ for almost every ω and every ψ of finite support. The latter are dense in $\ell^2(\mathbb{Z}^d)$ and thus $P_{\text{cont}}(h_\omega) \chi_I(h_\omega) = 0$ almost surely, meaning that the spectrum in I is pure point. $\square$

Chapter 3

Fractional moments

In this section we are going to see the important lemmas and theorems that help us proving localization. In particular we see the power of the fractional moment method, specifically on the *a priori bound*.

3.1 The Green's function

The *resolvent operator* is defined as:

$$G_z := (H_\omega - z)^{-1} \quad (3.1.1)$$

and is a well defined bounded operator for all $z \in \mathbb{C} \setminus \sigma\mathbb{R}$.

3.2 Localisation at large disorder

The goal of this thesis is to prove the following theorem.

Theorem 3.2.1 (Aizenman–Molchanov and Schenker). *Let $0 < s < 1$. Then there exists $\lambda_0 > 0$ such that for $\lambda > \lambda_0$ there are $C < \infty$ and $\mu > 0$ with*

$$\mathbb{E}(|G_z(x, y)|^s) \leq C e^{-\mu|x-y|} \quad (3.2.1)$$

uniformly in $x, y \in \mathbb{Z}^d$ and $z \in \mathbb{C} \setminus \mathbb{R}$. In particular, λ_0 is the unique solution of $\lambda_0 = \mu_d e \ln(\lambda_0)$ and for all $\lambda > \lambda_0 > e$ and $\epsilon > 0$ we have

$$\mathbb{E}\left(|G_z(x, y)|^{1 - \frac{1}{\ln(\lambda)}}\right) \leq \beta(\lambda) K_\epsilon \ln(\lambda) e^{-m_\epsilon(\lambda)|x-y|} \quad (3.2.2)$$

where

$$m_\epsilon(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

Remark 3.2.2. The reason why we distinguish (3.2.1) from (3.2.2) in this Theorem, is that the first one, represents a generalised bound for any absolutely continuous distribution of ω , while the second, considers the specific case of the uniform distribution between $[-1, 1]$, giving a more precise value of λ_0 . Both are equally valid to prove localization, but our goal will be to reach (3.2.2).

Before achieving this result, we start with two new lemmas, the so called *a priori bound* on the fractional moments and the resolvent identity.

Lemma 3.2.3 (A priori bound). *There exists a constant $C_1 = C_1(s) < \infty$ such that*

$$\mathbb{E}_x(|G_z(x, x)|^s) \leq C_1 \lambda^{-s} \quad (3.2.3)$$

for all $x \in \mathbb{Z}^d$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\lambda > 0$. In particular, for $\omega_x \sim \mathcal{U}[-1, 1]$

$$C_1 = \frac{1}{1-s}$$

Lemma 3.2.4 (Resolvent identity). *For a linear operator $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ and $z_1, z_2 \in \varrho(A)$,*

$$(A - z_1)^{-1} - (A - z_2)^{-1} = (A - z_1)^{-1}(z_1 - z_2)(A - z_2)^{-1}$$

and moreover, if $B : \mathcal{D}(B) \rightarrow \mathcal{H}$ is defined on a common domain, $\mathcal{D}(A) = \mathcal{D}(B)$,

$$(A - z)^{-1} - (B - z)^{-1} = (A - z)^{-1}(B - A)(B - z)^{-1}$$

for any $z \in \varrho(A) \cap \varrho(B)$.

The resolvent identity comes from a simple development of the first subtraction

Proof of a priori bound. This proof demonstrates the simplicity of the fundamental idea underlying the FMM. For fixed $x \in \mathbb{Z}^d$, write $\omega = (\hat{\omega}, \omega_x)$ where $\hat{\omega}$ is short for $(\omega_n)_{n \in \mathbb{Z}^d \setminus \{x\}}$. With $P_{e_x} := \langle e_x, \cdot \rangle e_x$, the orthogonal projection onto the span of e_x , we can separate the ω_x and $\hat{\omega}$ dependence of H_ω as

$$H_\omega = -\Delta + \lambda V_{\hat{\omega}} + \lambda \omega_x P_{e_x} = H_{\hat{\omega}} + \lambda \omega_x P_{e_x}.$$

For the operators $(H_\omega - z)$ and $(H_{\hat{\omega}} - z)$, the resolvent identity yields

$$(H_\omega - z)^{-1} = (H_{\hat{\omega}} - z)^{-1} - \lambda \omega_x (H_{\hat{\omega}} - z)^{-1} P_{e_x} (H_\omega - z)^{-1} \quad (3.2.4)$$

Taking matrix-elements we conclude for the corresponding diagonal Green functions that

$$G_z(x, x) = \hat{G}_z(x, x) - \lambda \omega_x \hat{G}_z(x, x) G_z(x, x) \quad (3.2.5)$$

where $\hat{G}_z$ denotes the Green's function of $H_{\hat{\omega}}$. Collecting in $G_z(x, x)$:

$$G_z(x, x) = \frac{1}{a + \lambda \omega_x} \quad \text{with } a = \frac{1}{\hat{G}_z(x, x)}. \quad (3.2.6)$$

Note that the latter is well-defined since one can easily check that $\text{Im} \hat{G}_z(x, x) > 0$ for $\text{Im} z > 0$.

The important fact is that a is a complex number which does not depend on ω_x . Thus, writing $\mathbb{E}_x(\dots) := \int \dots \rho(\omega_x) d\omega_x$ for $\rho(\omega_x) = \frac{1}{2} \chi_{[-1,1]}$, we find that

$$\mathbb{E}_x(|G_z(x, x)|^s) = \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{|\frac{a}{\lambda} + \omega_x|^s} \leq \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{|\omega_x|^s} = \frac{1}{(1-s)\lambda^s}, \quad (3.2.7)$$

with $\frac{1}{1-s}$ independent of λ and a , and thus independent of $\hat{\omega}$, z and x . □

3.2.1 From fractional moment's exponential bound to dynamical localisation

In this subsection we aim to prove the following theorem:

Theorem 3.2.5. *If there exist $s \in (0, 1)$, $C < \infty$ and $\mu > 0$ such that*

$$\mathbb{E}(|G_{E+i\eta}(x, y)|^s) \leq C e^{-\mu|x-y|} \quad (3.2.8)$$

uniformly in $E \in \mathbb{R}$ and $\eta > 0$, then exponential localisation holds.

Before proving this theorem, we need two more lemmas.

Lemma 3.2.6. *For a bounded and self-adjoint operator A ,*

$$\|(A - z)^{-1}\| \leq \frac{1}{\text{Im}(z)} \quad (3.2.9)$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$.

Proof. For the definition of the operator norm, we know that

$$\|(A - z)^{-1}\| := \sup_{\mu \in \sigma((A-z)^{-1})} |\mu|$$

Let's consider a $\lambda \in \mathbb{C}$ and write $\mu = \frac{1}{\lambda - z}$. This means:

$$\begin{aligned} \frac{1}{\lambda - z} \in \sigma((A - z)^{-1}) &\iff (A - z)^{-1} - (\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - z)^{-1}(A - \lambda)(\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - \lambda) \text{ not invertible} \\ &\iff \lambda \in \sigma(A) \end{aligned}$$

Where the resolvent identity has been used in third Implication.

Finally we obtain:

$$\|(A - z)^{-1}\| = \sup_{\lambda \in \sigma(A)} \frac{1}{|\lambda - z|} = \frac{1}{\inf_{\lambda \in \sigma(A)} |\lambda - z|} = \frac{1}{\text{dist}(\sigma(A), z)}$$

For a bounded and self-adjoint operator, we know that the spectrum is real-valued. For $z \in \mathbb{C} \setminus \mathbb{R}$ the minimum distance between z and $\sigma(A)$ is the imaginary component of z :

$$\|(A - z)^{-1}\| \leq \frac{1}{|\text{Im}(z)|} \quad (3.2.10)$$

□

The next proposition shows one of the most important correlation between the fractional and second moments.

Proposition 3.2.7. *For every $s \in (0, 1)$ there exists a constant $C_2 := C_2(s, \rho) < \infty$ such that*

$$|\text{Im } z| \mathbb{E}_x (|G_z(x, y)|^2) \leq C_2 \mathbb{E}_x (|G_z(x, y)|^s) \quad (3.2.11)$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$ and $x, y \in \mathbb{Z}^d$.

Remark 3.2.8. As it has been explained in the Section 1.3, we prefer to use fractional moments, rather than directly prove localization from the squared power.

Proof. As in the proof of 3.2.3 we consider the Hamiltonian

$$H^{(\alpha)} = H_\omega + \alpha P_{e_x}.$$

Its Green function will be denoted by $G^{(\alpha)}$. Applying the resolvent identity,

we find

$$(H_\omega - z)^{-1} = (H^{(\alpha)} - z)^{-1} + \alpha(H_\omega - z)^{-1}P_{e_x}(H^{(\alpha)} - z)^{-1}$$

and

$$\begin{aligned} G_z(x, y) &= G_z^{(\alpha)}(x, y) + \alpha G_z(x, y)G^{(\alpha)}(x, x) \\ \Rightarrow G_z^{(\alpha)}(x, y) &= \frac{G_z(x, y)}{1 + \alpha G_z(x, x)} \\ &= \frac{1}{\alpha + G_z(x, x)^{-1}} \cdot \frac{G_z(x, y)}{G_z(x, x)}. \end{aligned} \quad (3.2.12)$$

For the special case $x = y$ and $\tilde{\alpha} = -\operatorname{Re} G_z(x, x)^{-1}$ we get from (3.2.12) and Lemma 3.2.6 that

$$\left| \frac{1}{\operatorname{Im} G_z(x, x)^{-1}} \right| = |G_z^{(\tilde{\alpha})}(x, x)| \leq \frac{1}{|\operatorname{Im} z|},$$

i.e. $|\operatorname{Im} G_z(x, x)^{-1}| \geq |\operatorname{Im} z|$. Inserting this into (3.2.12) gives

$$|\operatorname{Im} z| |G_z^{(\alpha)}(x, y)|^2 \leq \frac{|\operatorname{Im} G_z(x, x)^{-1}|}{|\alpha + G_z(x, x)^{-1}|^2} \cdot \frac{|G_z(x, y)|^2}{|G_z(x, x)|^2}. \quad (3.2.13)$$

On the other hand, we can bound the same expression by

$$\begin{aligned} |\operatorname{Im} z| |G_z^{(\alpha)}(x, y)|^2 &\leq |\operatorname{Im} z| \sum_{y' \in \mathbb{Z}^d} |G_z^{(\alpha)}(x, y')|^2 \\ &= |\operatorname{Im} z| \langle e_x (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1} e_x \rangle \end{aligned}$$

For H_ω bounded and self adjoint, also $H^{(\alpha)}$ is bounded and self adjoint, da $\alpha \in \mathbb{R}$ and $\alpha < \infty$. It holds then,

$$|\operatorname{Im} z| \langle e_x (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1} e_x \rangle = |\operatorname{Im} z| \langle e_x, (H^{(\alpha)} - \bar{z})^{-1} (H^{(\alpha)} - z)^{-1} e_x \rangle$$

Applying the resolvent identity:

$$(H^{(\alpha)} - \bar{z})^{-1} - (H^{(\alpha)} - z)^{-1} = (H^{(\alpha)} - \bar{z})^{-1} (\operatorname{Im}(z)) (H^{(\alpha)} - z)^{-1}$$

. We obtain:

$$\begin{aligned}
& |\operatorname{Im} z| \langle e_x, (H^{(\alpha)} - \bar{z})^{-1} (H^{(\alpha)} - z)^{-1} e_x \rangle \\
&= |\operatorname{Im} z| \langle e_x, \frac{1}{\operatorname{Im}(z)} \left[(H^{(\alpha)} - z)^{-1} - (H^{(\alpha)} - \bar{z})^{-1} \right] e_x \rangle \\
&= |\operatorname{Im} G^{(\alpha)}(x, x; z)| \\
&= \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2}, \tag{3.2.14}
\end{aligned}$$

where the last step used (3.2.12) with $x = y$.

For $t \geq 0$ one has $\min(1, t^2) \leq t^s$, with $s \in (0, 1)$. Using this to interpolate between (3.2.13) and (3.2.14) we get

$$|\operatorname{Im} z| |G^{(\alpha)}(x, y; z)|^2 \leq \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2} \cdot \frac{|G_\omega(x, y; z)|^s}{|G_\omega(x, x; z)|^s}. \tag{3.2.15}$$

We will now use the following “re-sampling trick”, which has the effect of creating an additional random variable (here α) to average over. For a non-negative Borel function f on $\mathbb{R}$,

$$\begin{aligned}
\iint f(\omega_x + \alpha) \rho(\omega_x + \alpha) d\alpha \rho(\omega_x) d\omega_x &= \iint f(\omega_x + \alpha) \rho(\omega_x + \alpha) \rho(\omega_x) d\omega_x d\alpha \\
&= \iint f(\omega_x) \rho(\omega_x) \rho(\omega_x - \alpha) d\omega_x d\alpha \\
&= \int f(\omega_x) \rho(\omega_x) \left(\int \rho(\omega_x - \alpha) d\alpha \right) d\omega_x \\
&= \int f(\omega_x) \rho(\omega_x) d\omega_x, \tag{3.2.16}
\end{aligned}$$

. where in the first and third steps we used Fubini, in the second we used translation invariance of Lebesgue measure and in the fourth we used the fact that $\int \rho(\omega_x - \alpha) d\alpha = 1$.

Choose $f(\omega_x) = |G_z(x, y)|^2$; then (3.2.15) and (3.2.16) yield

$$\begin{aligned}
|\operatorname{Im} z| \mathbb{E}_x(|G_z(x, y)|^2) &= |\operatorname{Im} z| \mathbb{E}_x \left(\int |G_z^{(\alpha)}(x, y)|^2 \rho(\omega_x + \alpha) d\alpha \right) \\
&\leq \mathbb{E}_x \left(|\operatorname{Im} G_z(x, x)^{-1}| \frac{|G_z(x, y)|^s}{|G_z(x, x)|^s} \int \frac{\rho(\omega_x + \alpha)}{|\alpha + G_z(x, x)^{-1}|^2} d\alpha \right). \tag{3.2.17}
\end{aligned}$$

We now use Lemma 3.2.9 below with $w = G_z(x, x)^{-1}$ to conclude

$$|\operatorname{Im} z| \mathbb{E}_x(|G_z(x, y)|^2) \leq C_2 \mathbb{E}_x(|G_z(x, y)|^s),$$

with a constant $C_2 < \infty$ which only depends on $\text{supp } \rho$, but not on x, y and z . $\square$

Lemma 3.2.9. *Let $0 < s < 1$. There exists a constant $C_2 := C_2(s) < \infty$ such that*

$$|\text{Im } w| \cdot |w|^s \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha \leq C_2$$

uniformly in $w \in \mathbb{C}$ and $\omega_x \sim \mathcal{U}[-1, 1]$.

Proof. The proof is that of [REFERENCE GUNTER STOLZ]. Since $|w|^s \leq |\alpha|^s + |\alpha + w|^s$, we bound:

1.

$$\begin{aligned} |\text{Im } w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha &\leq |\text{Im } w| \int \frac{\| |\alpha|^s \rho(\omega_x + \alpha) \|_\infty}{|\alpha + w|^2} d\alpha \\ &\leq |\text{Im } w| \| |\alpha|^s \rho(\omega_x + \alpha) \|_\infty \int \frac{1}{(\alpha + \text{Re } w)^2 + \text{Im } w^2} d\alpha \end{aligned}$$

for $a, b \in \mathbb{R}$ we have

$$\int \frac{1}{a^2 + b^2} da = \frac{1}{b} \left[\arctan \left(\frac{a}{b} \right) \right]_{-\infty}^{\infty} = \frac{\pi}{b}.$$

Thus for $\rho = \frac{1}{2} \chi_{[-1, 1]}$ and $|\alpha| \leq 1 + |\omega| \leq 2$

$$\begin{aligned} |\text{Im } w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha &\leq \pi \| |\alpha|^s \rho(\omega_x + \alpha) \|_\infty \\ &\leq \pi 2^{s-1}. \end{aligned}$$

2. Now, we bound the second expression.

(a) For $|\text{Im } w|$ big:

Since $|\alpha + w| \geq |\text{Im } w|$ and $\int \rho(\omega_x + \alpha) d\alpha = 1$:

$$I_2 \leq |\text{Im } w| \cdot \frac{1}{|\text{Im } w|^{2-s}} \int \rho(\omega_x + \alpha) d\alpha = \frac{1}{|\text{Im } w|^{1-s}}$$

(b) For $|\text{Im } w|$ small:

$$\begin{aligned}
|\operatorname{Im} w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}} d\alpha &\leq |\operatorname{Im} w| \frac{1}{2} \int \frac{d\alpha}{|\alpha + w|^{2-s}} \\
&= |\operatorname{Im} w| \frac{1}{2} \int \frac{1}{|u + \operatorname{Im} w|^{2-s}} du \\
&= |\operatorname{Im} w|^2 \frac{1}{2} \int \frac{1}{|u^2 + \operatorname{Im} w^2|^{\frac{2-s}{2}}} du \\
&= \frac{1}{2} (\operatorname{Im} w)^s \int \frac{1}{|t^2 + 1|^{\frac{2-s}{2}}} dt
\end{aligned}$$

by setting $u = \alpha + \operatorname{Re} w$ and $t = \frac{u}{\operatorname{Im} w}$.

To establish the finiteness of the constant, we consider the integral

$$C_s := \int_{-\infty}^{\infty} \frac{1}{(t^2 + 1)^{\frac{2-s}{2}}} dt.$$

For $s \in (0, 1)$, the exponent satisfies $2 - s > 1$, which ensures the asymptotic decay $(t^2 + 1)^{-\frac{2-s}{2}} \sim |t|^{-(2-s)}$ as $|t| \rightarrow \infty$, rendering the integral absolutely convergent. Then we have

$$|\operatorname{Im} w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}} d\alpha \leq \min \left(\frac{1}{|\operatorname{Im} w|^{1-s}}, C_s \frac{1}{2} |\operatorname{Im} w|^s \right) \leq C'_s 2^{s-1}.$$

□

Now we can proceed with the proof of the theorem. The proof, recalls the one proposed in [Sto11]

Proof of Theorem 3.2.5. For H_ω bounded and self-adjoint, from the spectral theorem (see Definition 1.2.1), it exists a complex Borel spectral measure $\mu_{\delta_x, \delta_y} = \mu_{x,y}$ such that:

$$\mu_{x,y}(B) := \langle \delta_x, \chi_B(H_\omega) \delta_y \rangle.$$

with B Borel in $\mathbb{R}$. The *total variation measure* $|\mu_{x,y}|$ is a regular bounded Borel measure, and satisfies

$$|\mu_{x,y}|(B) = \sup \sum_{i=1}^{\infty} |\mu_{x,y}|(B_j) \quad (3.2.18)$$

where $B = \bigcup B_j$, with B_j disjoint. For $\mu_{x,y}$ complex, there exists α_j complex, such that $|\mu_{x,y}(B_j)| = \alpha_j \mu_{x,y}(B_j)$ and $|\alpha_j| = 1$. Let $g := \sum_{j=1}^{\infty} \alpha_j \chi_{B_j}$, which

implies that $|g| \leq 1$. Then it holds:

$$\left| \int_B g(\xi) d\mu_{x,y}(\xi) \right| = \left| \sum_{j=1}^{\infty} \alpha_j \mu_{x,y}(B_j) \right| = \sum_{j=1}^{\infty} |\mu_{x,y}(B_j)|. \quad (3.2.19)$$

From 3.2.18 we obtain:

$$\begin{aligned} |\mu_{x,y}|(B) &= \sup_{\substack{g: \mathbb{R} \rightarrow \mathbb{C} \text{ Borel} \\ |g| \leq 1}} \left| \int_B g(\xi) d\mu_{x,y}(\xi) \right| \\ &= \sup_{|g| \leq 1} |\langle \delta_x, g(H_\omega) \chi_B(H_\omega) \delta_y \rangle|, \end{aligned} \quad (3.2.20)$$

e.g. [Rud87, Chapter 6]. This implies for $g = e^{-itH_\omega}$:

$$\sup_{t \in \mathbb{R}} |\langle \delta_x, e^{-itH_\omega} \chi_B(H_\omega) \delta_y \rangle| \leq |\mu_{x,y}|(B).$$

Therefore, Theorem 3.2.5 will follow from a corresponding exponential decay bound for $\mathbb{E}(|\mu_{x,y}|(B))$.

From Lusin's theorem, we can extend the bound also for $g \in C_c(B)$:

$$|\mu_{x,y}|(B) = \sup_{\substack{g \in C_c(B) \\ |g| \leq 1}} |\langle \delta_x, g(H_\omega) \delta_y \rangle|.$$

for B bounded open interval. For $g \in C_c(B)$ it follows (using that g is bounded and uniformly continuous) that, uniformly in $\xi \in \mathbb{R}$,

$$g(\xi) = \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int \frac{g(E)}{(\xi - E)^2 + \eta^2} dE.$$

Now, always assuming that $g \in C_c(B)$, by the spectral theorem (see Def.

1.2.1), dominated convergence and Fubini:

$$\begin{aligned}
\langle \delta_x, g(H_\omega) \delta_y \rangle &= \int_{\mathbb{R}} g(\xi) d\mu_{x,y}(\xi) \\
&= \lim_{\eta \downarrow 0} \int_{\mathbb{R}} \left(\frac{\eta}{\pi} \int_B \frac{g(E)}{(\xi - E)^2 + \eta^2} dE \right) d\mu_{x,y}(\xi) \\
&= \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_B g(E) \left(\int_{\mathbb{R}} \frac{1}{(\xi - E)^2 + \eta^2} d\mu_{x,y}(\xi) \right) dE \\
&= \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_B g(E) \langle \delta_x, ((H_\omega - E)^2 + \eta^2)^{-1} \delta_y \rangle dE \\
&= \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_B g(E) \langle \delta_x, (H_\omega - E - i\eta)^{-1} (H_\omega - E + i\eta)^{-1} \delta_y \rangle dE
\end{aligned}$$

Now we can bound $\mathbb{E}(|\mu_{x,y}|)$, considering that $|g(E)| \leq 1$:

$$\begin{aligned}
\mathbb{E}(|\mu_{x,y}|(B)) &\leq \mathbb{E} \left(\lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_B g(E) \langle \delta_x, (H_\omega - E - i\eta)^{-1} (H_\omega - E + i\eta)^{-1} \delta_y \rangle dE \right) \\
&\leq \mathbb{E} \left(\lim_{\eta \rightarrow 0^+} \frac{\eta}{\pi} \int_B \sum_{z \in \mathbb{Z}^d} |\langle \delta_x, (H_\omega - E - i\eta)^{-1} \delta_z \rangle| |\langle \delta_z, (H_\omega - E + i\eta)^{-1} \delta_y \rangle| dE \right) \\
&\leq \lim_{\eta \rightarrow 0^+} \frac{1}{\pi} \int_B \sum_{z \in \mathbb{Z}^d} (\mathbb{E}(\eta |\langle \delta_x, (H_\omega - E - i\eta)^{-1} \delta_z \rangle|^2))^{1/2} \\
&\quad \cdot (\mathbb{E}(\eta |\langle \delta_z, (H_\omega - E + i\eta)^{-1} \delta_y \rangle|^2))^{1/2} dE,
\end{aligned}$$

where, in this order, Fatou, Fubini and Cauchy-Schwarz (on $\mathbb{E}$) have been used. Now we can use Proposition 3.2.7 and the hypothesis to obtain:

$$\begin{aligned}
\mathbb{E}(|\mu_{x,y}|(B)) &\leq \frac{C_1}{\pi} \int_B \sum_{z \in \mathbb{Z}^d} (\mathbb{E}|G_\omega(x, z; E + i\eta)|^s)^{1/2} (\mathbb{E}|G_\omega(z, y; E - i\eta)|^s)^{1/2} dE \\
&\leq \frac{C_1|B|C}{\pi} \sum_{z \in \mathbb{Z}^d} e^{-\frac{\mu}{2}|x-z|} e^{-\frac{\mu}{2}|z-y|}.
\end{aligned}$$

By the triangle inequality, $\frac{\mu}{2}|x-z| + \frac{\mu}{2}|z-y| \geq \frac{\mu}{4}|x-y| + \frac{\mu}{4}|x-z| + \frac{\mu}{4}|z-y|$. Therefore,

$$\mathbb{E}(|\mu_{x,y}|(B)) \leq \left(\frac{C_1|B|C}{\pi} \sum_{z \in \mathbb{Z}^d} e^{-\frac{\mu}{4}|x-z|} e^{-\frac{\mu}{4}|z-y|} \right) e^{-\frac{\mu}{4}|x-y|} \leq \tilde{C} e^{-\tilde{\mu}|x-y|},$$

with $\tilde{\mu} = \frac{\mu}{4}$ and finite prefactor $\tilde{C} = \frac{C_1|B|C}{\pi} \sum_{u \in \mathbb{Z}^d} e^{-\frac{\mu}{4}|u|}$. $\square$

From (2.2.2) we can also directly obtain spectral localisation through Simon-Wolff criterion.

Proposition 3.2.10 (Simon-Wolff criterion). *Let H_ω be a random operator in $l^2(\mathbb{Z}^d)$ with a random potential whose conditional-single site distribution is absolutely continuous. Then if for Lebesgue-almost every $E \in \mathbb{R}$ and $\mathbb{P}$ -almost all ω :*

$$\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 < \infty \quad (3.2.21)$$

H_ω has only pure point spectrum.

Proof can be found in [AW15b, Chapter 5]

Theorem 3.2.11. *Let Theorem 3.2.1 hold. Then, the spectrum of H_ω consists only of pure point spectrum for a.e. realisation of V_ω .*

Proof from [DERM24]. From (3.2.1) we notice that for all $\lambda > \lambda_{\text{And}}$:

$$\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x, y)|^s] = C \sum_{y \in \mathbb{Z}^d} e^{-\mu|x-y|} < \infty \quad \text{for all } \mu > 0 \quad (3.2.22)$$

uniformly for $z \in \mathbb{C} \setminus \mathbb{R}$ and $x \in \mathbb{Z}^d$ fixed.

From Proposition 3.2.7 we notice also that $\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x, y)|^2]$ always exists but might be infinite for $\eta \rightarrow 0$. We argue:

$$\begin{aligned} \mathbb{E} \left[\left(\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 \right)^{\frac{s}{2}} \right] &= \mathbb{E} \left[\lim_{\eta \downarrow 0} \left(\sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 \right)^{\frac{s}{2}} \right] \\ &\leq \mathbb{E} \left[\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^s \right] \\ &\leq \liminf_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x, y)|^s] < \infty, \end{aligned}$$

where in the first two steps we used that the function $x \mapsto x^{\frac{s}{2}}$ is monotone and $|\sum_n a_n|^s \leq \sum_n |a_n|^s$ for all $a_n \geq 0$ and $0 < s < 1$. The last two inequalities follow by Fatou and (3.2.22). $\square$

Chapter 4

SAW representation

4.1 SAW Definitions

Definition 4.1.1 (Self-Avoiding Walk). A **self-avoiding walk** on the integer lattice $\mathbb{Z}^d$ of length $N \in \mathbb{N}_0$ is an ordered $(N + 1)$ -tuple of distinct lattice sites

$$\gamma = (\gamma_0, \gamma_1, \dots, \gamma_N) \in (\mathbb{Z}^d)^{N+1}$$

such that consecutive vertices are nearest neighbors, that is,

$$|\gamma_i - \gamma_{i-1}| = 1 \quad \text{for all } i \in \{1, \dots, N\},$$

and $\gamma_i \neq \gamma_j$ for all $i \neq j$.

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Definition 4.1.2 (Sets of Self-Avoiding Walks). Let $\Lambda \subseteq \mathbb{Z}^d$. For any $x, y \in \Lambda$ and $N \in \mathbb{N}_0$:

- (i) $\mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N contained entirely in Λ , starting at $\gamma_0 = x$ and terminating at $\gamma_N = y$. When $\Lambda = \mathbb{Z}^d$, we simply write $\mathcal{S}_N(y, x)$.
- (ii) $\mathcal{S}_N^{(\Lambda)}(x) := \bigcup_{y \in \Lambda} \mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N in Λ starting at x .
- (iii) $c_N := |\mathcal{S}_N(x)| = \sum_{y \in \mathbb{Z}^d} |\mathcal{S}_N(y, x)|$ denotes the total number of self-avoiding walks of length N on $\mathbb{Z}^d$ originating from a fixed site (e.g., the origin 0), which is independent of the starting vertex by translation invariance.

Definition 4.1.3 (Connectivity Constant). The **connective constant** (or connectivity constant) μ_d of the lattice $\mathbb{Z}^d$ is defined as the asymptotic growth rate of c_N :

$$\mu_d := \lim_{N \rightarrow \infty} (c_N)^{1/N}.$$

By subadditivity of $\log c_N$, this limit exists, is positive, and satisfies the strict inequality $\mu_d < 2d - 1$.

Claim 4.1.4. Let μ_d be the connective constant (or growth rate) of self-avoiding walks on $\mathbb{Z}^d$. Then:

$$\mu_d \leq 2d - 1$$

Proof. Let c_N denote the number of self-avoiding walks (SAWs) of length N starting from the origin $0 \in \mathbb{Z}^d$, and let $S_N(0, x)$ denote the set of such walks ending at $x \in \mathbb{Z}^d$. By definition:

$$c_N = \sum_{x \in \mathbb{Z}^d} |S_N(0, x)| = \sum_{x \in \mathbb{Z}^d} \sum_{\gamma \in S_N(0, x)} 1$$

Consider the generating function (*susceptibility*) with parameter $\xi \in \mathbb{C}$:

$$\sum_{N=0}^{\infty} |\xi|^N c_N = \sum_{N=0}^{\infty} |\xi|^N \sum_{x \in \mathbb{Z}^d} |S_N(0, x)|$$

For $N = 0$, $c_0 = 0$. For $N \geq 1$, a walk cannot immediately step back onto the previous vertex (non-reversing condition). Thus:

- The first step has at most $2d$ choices.
- Each of the subsequent $N - 1$ steps has at most $2d - 1$ choices.

This gives the upper bound:

$$c_N \leq 2d(2d - 1)^{N-1}, \quad \text{for } N \geq 1.$$

Using this upper bound, we can bound the series from above:

$$\begin{aligned}
 \sum_{N=0}^{\infty} |\xi|^N c_N &\leq \sum_{N=1}^{\infty} |\xi|^N (2d)(2d-1)^{N-1} \\
 &= 2d|\xi| \sum_{k=0}^{\infty} |\xi|^k (2d-1)^k \quad (\text{setting } k = N-1) \\
 &= 2d|\xi| \sum_{k=0}^{\infty} (|\xi|(2d-1))^k
 \end{aligned}$$

This geometric series converges if and only if:

$$|\xi|(2d-1) < 1 \iff |\xi| < \frac{1}{2d-1} =: R$$

where $R = \frac{1}{2d-1}$ is the radius of convergence of the bounding series.

By the Cauchy–Hadamard theorem, the radius of convergence R' of the original series $\sum_{N=0}^{\infty} c_N |\xi|^N$ is:

$$R' = \frac{1}{\limsup_{N \rightarrow \infty} c_N^{1/N}} = \frac{1}{\mu_d}$$

Since the original series is bounded term-by-term by a series that converges for $|\xi| < R$, the true radius of convergence must satisfy:

$$R' \geq R \implies \frac{1}{\mu_d} \geq \frac{1}{2d-1}$$

Taking the reciprocal yields:

$$\mu_d \leq 2d-1.$$

□

Definition 4.1.5 (SAW Correlation Function). For a parameter $\beta \geq 0$, the **self-avoiding walk correlation function** $C_\beta(y-x)$ between two vertices $x, y \in \mathbb{Z}^d$ is the generating function

$$C_\beta(y-x) := \sum_{N=0}^{\infty} \beta^N |\mathcal{S}_N(y, x)|,$$

defined for all values of β for which the power series is absolutely convergent. By translation invariance, $C_\beta(y-x)$ depends only on the displacement vector

$y - x$.

Definition 4.1.6 (SAW Susceptibility). The **susceptibility** $\chi(\beta)$ of self-avoiding walks is the sum of the two-point correlation function over all endpoints:

$$\chi(\beta) := \sum_{x \in \mathbb{Z}^d} C_\beta(x) = \sum_{N=0}^{\infty} c_N \beta^N.$$

The series converges uniformly and absolutely if and only if $(c_N)^{1/N} \beta < 1$, where the critical radius of convergence R_χ is given by $\lim_{N \rightarrow \infty} \frac{1}{(c_N)^{1/N}} = \frac{1}{\mu_d}$.

Proposition 4.1.7 (Exponential Decay of the Correlation Function). *Let $\beta \geq 0$ be such that $\beta < 1/\mu_d$. Then for every $\varepsilon > 0$ there exists a constant $K_\varepsilon < \infty$, independent of x , such that*

$$C_\beta(x) \leq K_\varepsilon ((\mu_d + \varepsilon)\beta)^{|x|} \quad \text{for all } x \in \mathbb{Z}^d.$$

Proof. We know that $|\mathcal{S}_N(0, x)| = 0$ whenever $N < |x|$, and the sum defining $C_\beta(x)$ may be restricted to $N \geq |x|$:

$$C_\beta(x) = \sum_{N=|x|}^{\infty} \beta^N |\mathcal{S}_N(x, 0)| \leq \sum_{N=|x|}^{\infty} \beta^N c_N. \quad (4.1.1)$$

Fix $\varepsilon > 0$. Since $\mu_d = \lim_{N \rightarrow \infty} c_N^{1/N}$, by the definition of limit there exists $N_0 \in \mathbb{N}$ such that

$$c_N^{1/N} < \mu_d + \varepsilon \quad \text{for all } N \geq N_0,$$

that is,

$$c_N < (\mu_d + \varepsilon)^N \quad \text{for all } N \geq N_0.$$

For $N = 0, 1, \dots, N_0 - 1$, define

$$K_\varepsilon := \max \left(1, \max_{0 \leq N < N_0} \frac{c_N}{(\mu_d + \varepsilon)^N} \right),$$

which is finite, being the maximum of finitely many finite quantities. By construction it holds:

$$c_N \leq K_\varepsilon (\mu_d + \varepsilon)^N \quad \text{for all } N \geq 0.$$

Combining it with (4.1) we obtain

$$C_\beta(x) \leq K_\varepsilon \sum_{N=|x|}^{\infty} [(\mu_d + \varepsilon)\beta]^N.$$

Choosing ε small enough that $r := (\mu_d + \varepsilon)\beta < 1$ (possible since $\beta < 1/\mu_d$), the tail of the geometric series equals

$$\sum_{N=|x|}^{\infty} r^N = \frac{r^{|x|}}{1-r}.$$

Hence

$$C_\beta(x) \leq \frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta} [(\mu_d + \varepsilon)\beta]^{|x|}.$$

where $\frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta}$ is still finite and independent from x . $\square$

4.2 SAW representation of G

The proof of the theorem 3.1 is similar to the one given by Schenker in [1]. We are going to work with the Green's function defined on a volume where one site is removed every stage. Therefore, for any $\Lambda \subseteq \mathbb{Z}^d$ (finite or infinite) the Green function changes:

$$G_z^\Lambda(x, y) := \begin{cases} (H_\omega^\Lambda - z), & \forall x, y \in \Lambda \\ 0 & , \text{otherwise} \end{cases} \quad (4.2.1)$$

where $H_\omega^\Lambda - z \in \mathbb{C}^{\Lambda \times \Lambda}$ is the matrix obtained by restricting $H_\omega - z$ to Λ . In this thesis we will consider $\Lambda = \mathbb{Z}^d$, but we leave the notation Λ to highlight the fact that the same result holds for any volume.

To notice is also that $H_\omega^\Lambda - z$ is still invertible for all $z \in \mathbb{C} \setminus \mathbb{R}$. In this section, for simplicity during calculations, we will write $G^\Lambda(x, y)$ instead of $G_z^\Lambda(x, y)$.

Theorem 4.2.1. *Let $x, y \in \mathbb{Z}^d$, $x \neq y$, $\Lambda \subseteq \mathbb{Z}^d$. G^Λ is defined as in (4.1). It holds*

$$G^\Lambda(x, y) = \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \prod_{i=1}^n G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) + R_N \quad (4.2.2)$$

where $\Lambda_i = \Lambda \setminus \{x, \gamma_1, \dots, \gamma_{i-1}\}$ and

$$R_N = \sum_{x \sim y} \sum_{\gamma \in \mathcal{S}_N(0, x)} \prod_{i=1}^{N-1} G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) \cdot G^{\Lambda_N}(\gamma_N, y). \quad (4.2.3)$$

In order to prove Theorem 4.8,

Claim 4.2.2.

$$G^\Lambda(x, y) = G^\Lambda(x, x) \sum_{x_1 \sim x} G^{\Lambda_1}(x_1, y) \quad (4.2.4)$$

Proof. Using the resolvent identity for $A = (H_\omega^\Lambda - z)$ and $B = (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)$, we obtain:

$$\begin{aligned} & (H_\omega^\Lambda - z)^{-1} - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \\ &= -(H_\omega^\Lambda - z)^{-1} \left(H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) \right) (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \end{aligned}$$

Moreover:

$$H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) = - \sum_{x_1 \sim x} \delta_{x_1} =: T_{x, x_1}$$

Now considering the matrix elements in (x, y) with $x \neq y$ and matrix multiplication:

$$\langle \delta_x, (H_\omega^\Lambda - z)^{-1} \delta_y \rangle - 0 = - \sum_{x_1, x_2} \langle \delta_x, (H_\omega^\Lambda - z)^{-1} \delta_{x_1} \rangle T_{x_1, x_2} \langle \delta_{x_2}, (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \delta_y \rangle$$

For $T_{x_1, x_2} \neq 0$, we need $x_1 = x$ and $x_2 \sim x_1$.

Which leaves us with:

$$G^\Lambda(x, y) = \sum_{x_1 \sim x} G^\Lambda(x, x) G^{\Lambda \setminus \{x\}}(x_1, y)$$

□

Proof of Theorem 5.8. Now we separate the case in which $x_1 = y$ (which implies $|x - y| = 1$).

$$G^\Lambda(x, y) = G^\Lambda(x, x) \left(\sum_{x_1 \sim x, x_1 \neq y} G^{\Lambda_1}(x_1, y) + \delta_{|x-y|=1} G^{\Lambda_1}(y, y) \right)$$

Iterating N times, it gives (4.2.2). □

4.3 Proof with SAW

In this last Section, we can use the SAW representation of the Green function (4.2.2), in order to prove Theorem 3.2.1.

Proof. We take the average of (4.2), and using $|\sum a|^s \leq \sum |a|^s$ (which follows from triangle inequality), we have:

$$\mathbb{E} [|G^\Lambda(x, y)|^s] \leq \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{i=1}^n |G^{\Lambda_i}(\gamma_i, \gamma_i)|^s \right] + \quad (4.3.1)$$

$$\sum_{x \sim y} \sum_{\gamma \in \mathcal{S}_N(0, x)} \mathbb{E} [|G^\Lambda(x, x)|^s \prod_{i=1}^{N-1} |G^{\Lambda_i}(\gamma_i, \gamma_i)|^s \cdot |G^{\Lambda_N}(\gamma_N, y)|^s] \quad (4.3.2)$$

The resolvent $G^{\Lambda_j}(w, w')$ does not depend on the random variables ω_{x_i} , $i = 0, \dots, j-1$. Hence, recursive applications of the a priori bound (3.2) yield

$$\mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^n |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s \right] \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^{n+1}$$

and

$$\begin{aligned} & \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^{N-1} |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s \cdot |G^{\Lambda_{N-1}}(\gamma_N, y)|^s \right] \\ & \leq \left(\frac{1}{1-s} \cdot \frac{1}{\lambda^s} \right)^N \cdot \mathbb{E} [|G^{\Lambda_{N-1}}(\gamma_N, y)|^s] \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^N \frac{1}{|\operatorname{Im}(z)|^s} \end{aligned}$$

Where we used a priori bound and lemma 3.5.

Inserting these estimates in the sums above, we get:

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=0}^N \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y, x)| + \operatorname{Err}(N) \frac{1}{|\operatorname{Im} z|^s} \quad (4.3.3)$$

with $\operatorname{Err}(N) := \beta(s)^N |\mathcal{S}_N^{(\Lambda)}(x)|$ $\beta(s) = \frac{1}{1-s} \frac{1}{\lambda^s}$ Let's consider the susceptab-

ility $\chi(\beta)$ of the self avoiding walks in $\mathcal{S}_N^{(\Lambda)}(x)$:

$$\chi(\beta) = \sum_{x \in \mathbb{Z}^d} \sum_{N=0}^{\infty} \beta(s)^N |S_N^{(\Lambda)}(x)| = \sum_{N=0}^{\infty} c_N \beta(s)^N. \quad (4.3.4)$$

The series converges absolutely if and only if $0 \leq |\beta| < 1/\mu_d$, where $1/\mu_d = \lim_{N \rightarrow \infty} c_N^{1/N}$ is the critical radius of convergence. Thus, for $|\beta(s)| = \frac{1}{|(1-s)\lambda^s|} \leq \frac{1}{\mu_d}$, $\chi(\beta)$ converges uniformly, and most importantly:

$$Err(N) \longrightarrow 0, \quad \text{for } N \rightarrow \infty$$

We send N to the infinity and obtain from (4.7):

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=0}^{\infty} \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y, x)| \quad (4.3.5)$$

Taking the supremum over volumes Λ now yields

$$\sup_{\Lambda \subset \mathbb{Z}^d} \mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \beta(s) C_{\beta(s)}(x - y). \quad (4.3)$$

It remains to establish for which λ we may find $s \in (0, 1)$ with $\beta(s) < 1/\mu_d$. As above, for $\lambda > e$ the unique critical point is $s_{\text{crit}} = 1 - 1/\ln \lambda$ and

$$\min_{s \in (0, 1)} \beta(s) = \beta(s_{\text{crit}}) = \frac{e \ln \lambda}{\lambda}$$

which is less than $1/\mu_d$ if and only if $\lambda > \lambda_{\text{And}}$. Finally we get:

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y; z)|^{1 - \frac{1}{\ln \lambda}} \right) \leq \beta(\lambda) K_{\epsilon} \ln \lambda e^{-m_{\epsilon}(\lambda)|x-y|} \quad (4.3.6)$$

where

$$m_{\epsilon}(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

□

Chapter 5

Connection to Long-Range SAW

The purpose of this chapter is to extend the self-avoiding walk representation of the resolvent, to random Hamiltonians with long-range hopping. In particular, we compare this setting with the previous model and apply the fractional moment method to study localization.

The difference lies in the kinetic operator, which is replaced by the fractional discrete Laplacian $(-\Delta)^\alpha$ with $0 < \alpha < 1$. This change leads to a polynomial decay bound for the averaged Green's function, in sharp contrast with the standard exponential decay.

This chapter is based on the work of M. Disertori, R.M. Escobar and C. Rojas-Molina in 2024 [DERM24].

5.1 The fractional Laplacian

The fractional kinetic operator also connects arbitrarily distant sites in the lattice, rather than just the first neighbors.

Definition 5.1.1. Let $-\Delta$ be defined as in Section 1.1. The fractional discrete Laplacian is $(-\Delta)^\alpha$ for $0 < \alpha < 1$

Just as the standard discrete Laplacian, $(-\Delta)^\alpha$ is bounded, self-adjoint and translation invariant. Its entries satisfy

$$(-\Delta)^\alpha(x, x) > 0 \quad \forall x \in \mathbb{Z}^d, \quad (-\Delta)^\alpha(x, y) < 0 \quad \forall x \neq y \text{ in } \mathbb{Z}^d$$

Moreover, its off-diagonal matrix elements exhibit polynomial decay: there exist constants $0 < c_{\alpha,d} \leq C_{\alpha,d} < \infty$ such that

$$\frac{c_{\alpha,d}}{|x-y|^{d+2\alpha}} \leq -(-\Delta)^\alpha(x,y) \leq \frac{C_{\alpha,d}}{|x-y|^{d+2\alpha}} \quad (5.1.1)$$

We notice that for $x \neq y$, the fractional Laplacian is never exactly equal to zero. This characteristic describes the long-jump behavior: now, the discrete Laplacian reaches every site of the grid, since the values are never zero, in contrast with the standard case, where the only value different from zero were on the diagonal and first neighbor sites.

Moreover, (5.1.1) confirms the boundness of the fractional operator for all $x \neq y$ and the summability of its kernel $\sum_{x \in \mathbb{Z}^d} |(-\Delta)^\alpha(0,x)| < \infty$. For the diagonal entries, it holds:

$$(-\Delta)^\alpha(x,x) = - \sum_{y \neq x} (-\Delta)^\alpha(x,y)$$

which confirms boundness also for $x = y$. Notice that for $\alpha = 1$, we can use the same sum and obtain $(-\Delta)(x,x) = - \sum_y (-\Delta)(x,y) = 2d$.

We can already prove the summability of the fractional kernel too:

Lemma 5.1.2. *For any $\alpha \in (0,1)$ and $s \in \left(\frac{d}{d+2\alpha}, 1\right]$,*

$$\sum_{x \in \mathbb{Z}^d} |(-\Delta)^\alpha(0,x)|^s < \infty \quad (5.1.2)$$

Proof. From (5.1.1):

$$\sum_{|x| \geq 1, x \in \mathbb{Z}^d} |(-\Delta)^\alpha(0,x)|^s \leq C_{\alpha,d}^s \sum_{|x| \geq 1, x \in \mathbb{Z}^d} \frac{1}{|x|^{(d+2\alpha)s}}.$$

By standard comparison of the lattice sum with an integral on $\mathbb{R}^d$:

$$\sum_{|x| \geq 1, x \in \mathbb{Z}^d} \frac{1}{|x|^p} \leq C_d \int_{|z| > 1} \frac{1}{|z|^p} dz.$$

Passing to spherical coordinates in $\mathbb{R}^d$, the integral reads

$$\begin{aligned} \int_{|z|>1} \frac{1}{|z|^p} dz &= |\mathbb{S}^{d-1}| \int_1^\infty \frac{r^{d-1}}{r^p} dr \\ &= |\mathbb{S}^{d-1}| \int_1^\infty r^{d-1-p} dr, \end{aligned}$$

where $|\mathbb{S}^{d-1}| = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ denotes the surface area of the unit sphere in $\mathbb{R}^d$.

The radial integral converges if and only if the power satisfies

$$d - 1 - p < -1 \iff p > d.$$

Substituting $p = s(d + 2\alpha)$, this condition becomes

$$s(d + 2\alpha) > d \iff s > \frac{d}{d + 2\alpha}.$$

Since $\alpha > 0$ implies $d + 2\alpha > d$; for every $s \in \left(\frac{d}{d+2\alpha}, 1\right]$, we conclude that

$$\sum_{x \neq 0} |(-\Delta)_{0,x}^\alpha|^s \leq C_{\alpha,d}^s \sum_{x \neq 0} \frac{1}{|x|^{s(d+2\alpha)}} < \infty.$$

□

5.2 The SAW Representation for Long-Range Resolvents

We consider the fractional Hamiltonian

$$H_{\alpha,\omega} = (-\Delta)^\alpha + \lambda V_\omega \quad \text{on } \ell^2(\mathbb{Z}^d), \quad (5.2.1)$$

where $0 < \alpha < 1$ and V_ω is a diagonal matrix with i.i.d. random entries defined as in Section 1.1. The Green's function $G_z = (H_{\alpha,\omega} - z)^{-1}$ is well-defined for all $z \in \mathbb{C} \setminus \mathbb{R}$, and its decay at strong disorder is closely related to the polynomial decay of the fractional Laplacian.

As before, in the long-range scenario, we use the geometric resolvent expansion to represent the fractional moments as a sum over self-avoiding paths. Since we are considering arbitrary sites on the lattice, we define the step weight kernel for the SAW as $D(x, y) := |(-\Delta)^\alpha(x, y)|^s$, which generates

a transition probability $p(x \rightarrow y) := \frac{D(x,y)}{\sum_{z \neq x} D(x,z)}$. This is well-defined for $s \in \left(\frac{d}{d+2\alpha}, 1\right)$ because the kernel is summable: $\sum_{x \neq 0} |(-\Delta)^\alpha(0, x)|^s < \infty$ (See (5.1.2)). From this essential condition, we develop the same strategy as for the standard Anderson resolvent. The new definitions for SAWs are all generated by D .

SAW definitions for the discrete Anderson model. The averaged fractional Green function satisfies

$$\mathbb{E}[|G_z(x, y)|^s] \leq \frac{K_0}{(1-s)\lambda^s} \frac{1}{|x-y|^{d+2\alpha_s}} \quad \forall x, y \in \mathbb{Z}^d, x \neq y, \quad (5.2.2)$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$, for all $\lambda > \lambda_0(s) := \left(\frac{1}{(1-s)R_{\chi_D}}\right)^{\frac{1}{s}}$. Here, R_{χ_D} is the radius of convergence of the susceptibility generated by $D(0, x)$, and $2\alpha_s := s(d+2\alpha) - d$. The constant $K_0 = K_0(d, \alpha, s)$ comes from the two-point function bound of the self-avoiding walk, and $\frac{1}{(1-s)\lambda^s}$ comes from the *a priori bound*.

As in the previous Chapter, we prove (5.2.2) using a SAW representation of $G_{\alpha, \omega}$.

Theorem 5.2.1. *Let $s \in (\frac{d}{d+2\alpha}, 1)$. Then, for all the above definitions, it holds*

$$\mathbb{E}[|G_{\alpha, \omega}(x, y)|^s] \leq \frac{1}{\lambda^s(1-s)} C_\beta^D(y-x) + \frac{1}{|\text{Im}(z)|^s} R_N \quad (5.2.3)$$

for all $x, y \in \mathbb{Z}^d$. Moreover,

$$R_N := \left(\frac{1}{(1-s)\lambda^s}\right)^N \sum_{\gamma \in S_{N+1}(x, y)} \prod_{j=0}^{N-1} D(\gamma_j, \gamma_{j+1}). \quad (5.2.4)$$

1. **Resolvent Identity Iteration:** We expand the restricted Green's function $G_z^\Lambda(x, y)$ by iterating the resolvent identity N times, pulling out the off-diagonal elements $\Delta^\alpha(\gamma_j, \gamma_{j+1}) = -(-\Delta)^\alpha(\gamma_j, \gamma_{j+1}) > 0$ along self-avoiding paths $\gamma \in \mathcal{S}_n(x, y)$. Since $0 < \alpha < 1$, we keep the Laplacian fractional instead of reducing it to nearest-neighbor hopping.
2. **Averaging and A Priori Bound:** We take the fractional moment of the expansion and apply subadditivity. Using the a priori bound to

decouple the diagonal resolvent terms, we obtain:

$$\mathbb{E}[|G_z^\Lambda(x, y)|^s] \leq \frac{\theta_s}{\lambda^s} C_\beta^D(y - x) + \frac{1}{|\operatorname{Im} z|^s} \operatorname{Err}(N), \quad (5.2.5)$$

where C_β^D is the SAW two-point function with fugacity $\beta = \frac{\theta_s}{\lambda^s}$. For $\lambda > \lambda_0(s)$, we have $\beta < R_{\chi_D}$, which ensures the susceptibility is finite and $\operatorname{Err}(N) \rightarrow 0$ as $N \rightarrow \infty$.

3. **Two-Point Function Decay:** Finally, by [Lemma 4 of PAPER DISERTORI], the SAW correlation function is bounded by the generating kernel's decay:

$$C_\beta^D(y - x) \leq K_0 \frac{1}{|y - x|^{d+2\alpha_s}}, \quad (5.2.6)$$

yielding the final polynomial bound (5.2.2).

5.3 The Localisation Criteria

A natural question that may arise is how localisation is proven without the exponential decay of the fractional moments of the Green's function. Due to the polynomial tail of the fractional Laplacian, exponential decay is not expected; instead, it is only possible to prove a weaker version of dynamical localisation where only moments $\beta < 2\alpha$ of the position operator are finite. Spectral localisation is proven via the Simon-Wolff theorem.

Lemma 5.3.1. *Let (5.2.2) hold uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ and for $\lambda > \lambda_0(s)$. It follows that:*

- (i) *The spectrum of $H_{\alpha, \omega}$ consists only of pure point spectrum for a.e. $\omega \in \Omega$.*
- (ii) *For a.e. $\omega \in \Omega$ and for all E in the almost sure spectrum, the corresponding normalized eigenfunction φ_E satisfies:*

$$|\varphi_E(y, \omega)|^2 \leq \frac{C_t(\omega)}{(1 + |y - x_E(\omega)|)^t} \quad \forall 0 < t < 2\alpha_s, \quad (5.3.1)$$

where $x_E(\omega)$ is a localization center and $C_t(\omega)$ is an integrable random variable.

- (iii) *For $0 < \beta < 2\alpha_s$, we have for a.e. $\omega \in \Omega$*

$$\sup_{t \in \mathbb{R}} \sum_{x \neq 0} |x|^\beta |e^{-itH_{\alpha, \omega}}(0, x)|^2 < \infty. \quad (5.3.2)$$

In particular, if $\alpha > \frac{1}{2}$, the first moment of the wave function initially localized at the origin and evolving under the action of $H_{\alpha,\omega}$ is finite.

Proof. (i) From (5.2.2), we verify the summability of the Green's function moments:

$$\sum_{x \in \mathbb{Z}^d} \mathbb{E}[|G_z(0, x)|^s |x|^t] \leq \frac{K_0 \theta_s}{\lambda^s} \sum_{x \neq 0} \frac{1}{|x|^{d+2\alpha_s-t}} < \infty \quad \forall t \in [0, 2\alpha_s), \quad (5.3.3)$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ and for all $\lambda > \lambda_0(s)$. Setting $t = 0$ and applying Fatou's lemma ensures that $\liminf_{\eta \downarrow 0} \sum_y |G_{E+i\eta}(x, y)|^2 < \infty$ a.s.. We then apply the Simon-Wolff criterion to obtain spectral localisation.

- (ii) The eigenfunction polynomial decay follows from the summability bound (5.3.3) and standard fractional moment arguments.
- (iii) The dynamical bounds follow from bounding the eigenfunction correlator $Q(x, 0, \Sigma)$, which is majorized by the integrated Green's function moments. Since $\mathbb{E}[Q(x, 0, \Sigma)] \leq \frac{C}{|x|^{d+2\alpha_s}}$, the sum $\sum_{x \neq 0} |x|^\beta \mathbb{E}[Q(x, 0, \Sigma)]$ converges for $\beta < 2\alpha_s$, yielding the result.

□

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